NeuralMed

A Multi-Disease Clinical Screening Platform Using Classical ML and Deep Transfer Learning

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Overview

Early detection of chronic diseases using ML reduces diagnostic delays and extends preventive care to underserved populations. NeuralMed unifies **five disease-screening modules** and a **deep learning CT classifier** into a single Flask web application with two clinical portals.

Patient Portal

→ Plain-language predictions with Gemini AI explanation

→ Input validation, graceful error handling

Doctor Portal NEW in D3

→ Model confidence percentage

→ Top-10 feature importance attribution

→ Per-feature clinical reference ranges

→ Gemini-2.5-Flash-Lite structured summary

→ Browser-native PDF export

D2 → D3 Improvements

→ 5-fold stratified cross-validation added

→ max_depth=10 to fix 100% overfit in RF/DT

→ Gemini API updated: gemini-pro → gemini-2.5-flash-lite

→ Typed exception handlers replace broad except

System Architecture

DATA INPUT

Tabular: 5 clinical CSV datasets | Image: CT scan upload

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PREPROCESSING

StandardScaler · LabelEncoder · Median imputation Image: Resize 224×224 · Normalize [0,1] · Augmentation

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ALGORITHM LAYER

Disease	Algorithm	Why Chosen
Diabetes	GaussianNB	Best CV; handles small continuous features
Heart	Naive Bayes	Strong on correlated cardiac markers
Parkinson's	RF depth=10	Ensemble; depth cap fixes D2 overfit
Breast Cancer	Logistic Reg.	Linear separability in FNA feature space
Kidney CT	VGG-16 TL	ImageNet weights; fine-tuned last layers

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OUTPUT LAYER

Patient Portal	Doctor Portal
Binary Prediction + Gemini Explanation	Confidence % · Feature Importance Reference Ranges · Clinical Summary

Algorithms & Evaluation

8 Classifiers Evaluated Per Disease

Naïve Bayes · Logistic Regression · k-NN · SVM (RBF) · Decision Tree · Random Forest · XGBoost · Extra Trees

5-Fold Stratified Cross-Validation

StratifiedKFold(n=5, shuffle=True, random_state=42) Reports: mean accuracy ± std, confusion matrix, ROC/AUC

Disease	Best Model	Test Acc	CV Mean	±Std	AUC
Diabetes	GaussianNB	74.7%	76.0%	1.7	0.84
Heart Disease	Naive Bayes	85.2%	84.1%	4.3	0.92
Parkinson's	RF depth=10	**92.3%**	88.2%	4.8	**0.97**
Breast Cancer	Logistic Reg.	**98.2%**	97.4%	1.7	**0.99**
Kidney CT	VGG-16 TL	**99.0%**	—	—	—

Overfitting Fix (D2 → D3)

D2: Random Forest + Decision Tree reached **100% train accuracy** (memorisation). D3 constrains **max_depth=10** — test accuracy maintained while CV gap closed.

CNN Architecture — Kidney CT

VGG-16 pretrained on ImageNet. Final dense layers replaced: GlobalAvgPool → Dense(256, ReLU) → Dropout(0.5) → Dense(4, Softmax). Classes: Normal · Cyst · Stone · Tumor. Dataset: 12,446 CT images, 80/10/10 split.

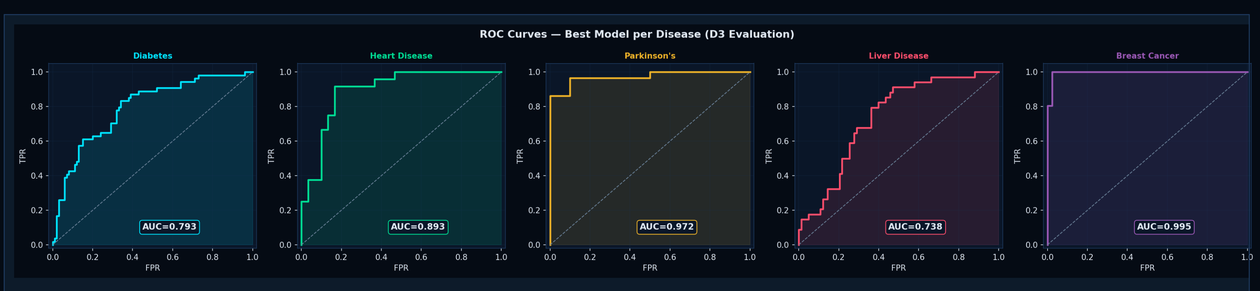


Fig. A. ROC Curves — AUC per disease

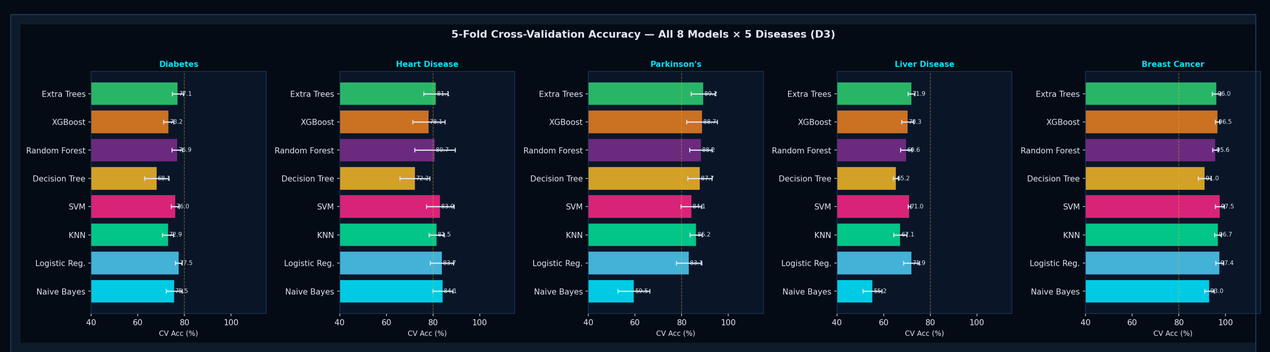
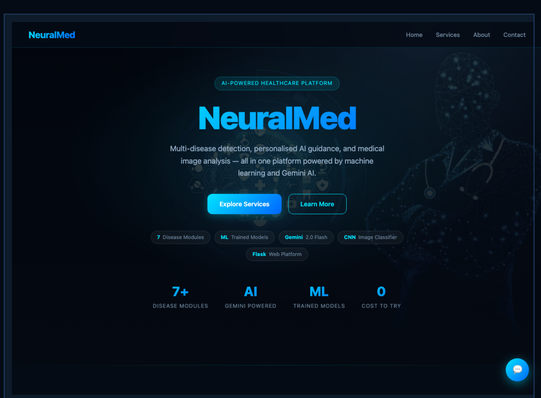
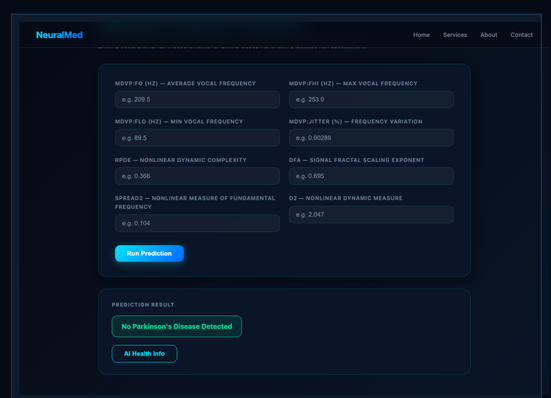


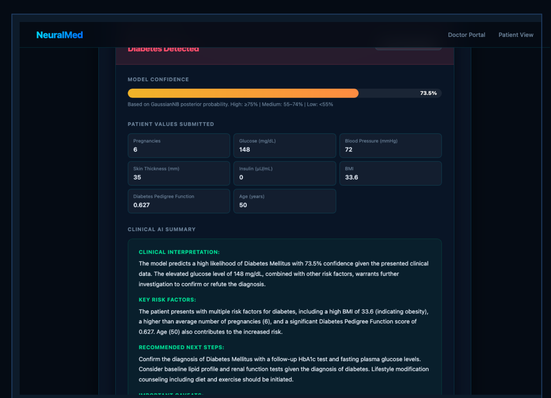
Fig. B. 5-Fold CV Accuracy — All Models × All Diseases



Landing — chatbot + modules



Parkinson's Detected — patient view



Doctor Portal — Diabetes Detected

Conclusion

What we built:

A production-ready Flask platform serving 5 disease-screening modules through dual clinical portals, with full evaluation pipeline.

Key results:

- Breast Cancer: 98.2% acc, AUC 0.99
- Kidney CT (VGG-16): 99.0% accuracy
- Parkinson's RF: 92.3% acc, 88.2% CV
- Heart Disease NB: 85.2% acc, AUC 0.92

Novel contributions:

- Unified multi-disease platform (5 modules, 2 portals)
- LLM-augmented clinical summaries via Gemini 2.5
- Depth-constrained ensemble with CV validation
- Doctor Portal with feature attribution + PDF export

Future work:

- Grad-CAM visualisations for CNN explainability
- SMOTE for class-imbalanced datasets
- Cloud deployment with ONNX runtime

github.com/honeyreddydaram/NeuralMed

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